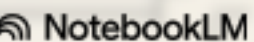
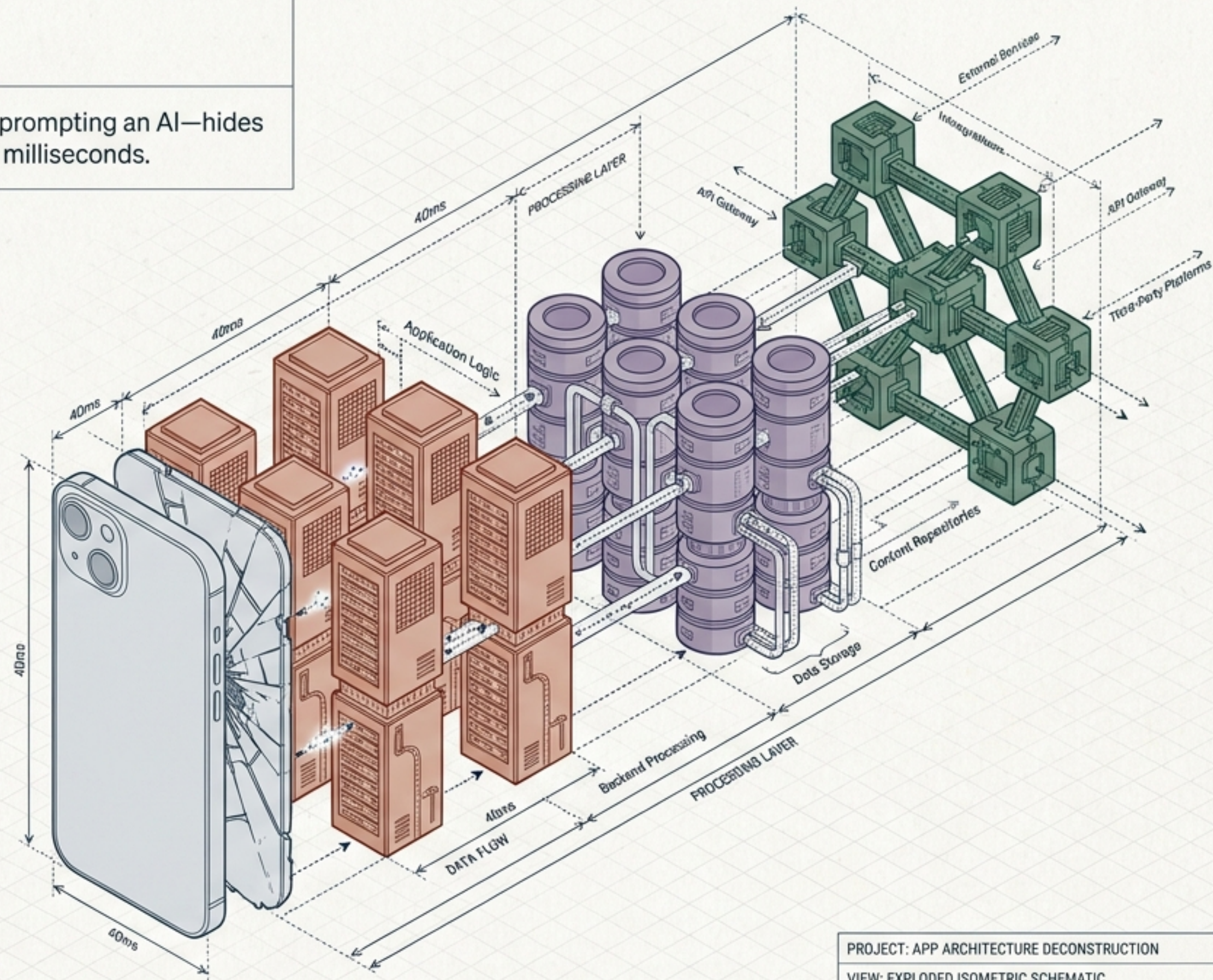
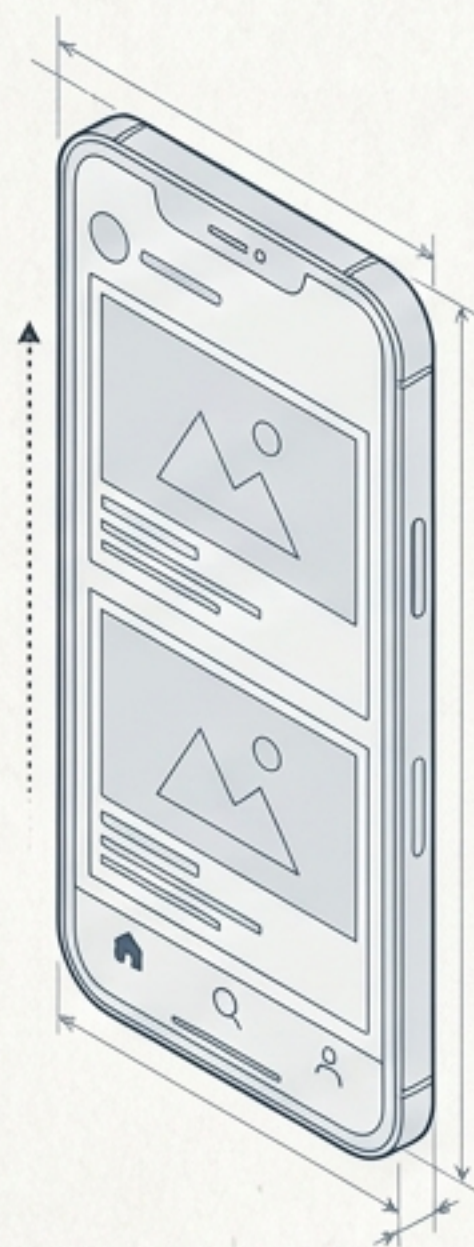


Week 1 • Topic 6



The Illusion of Simplicity

Every seamless app interaction—from scrolling a feed to prompting an AI—hides an invisible, synchronised network of systems working in milliseconds.



PROJECT: APP ARCHITECTURE DECONSTRUCTION

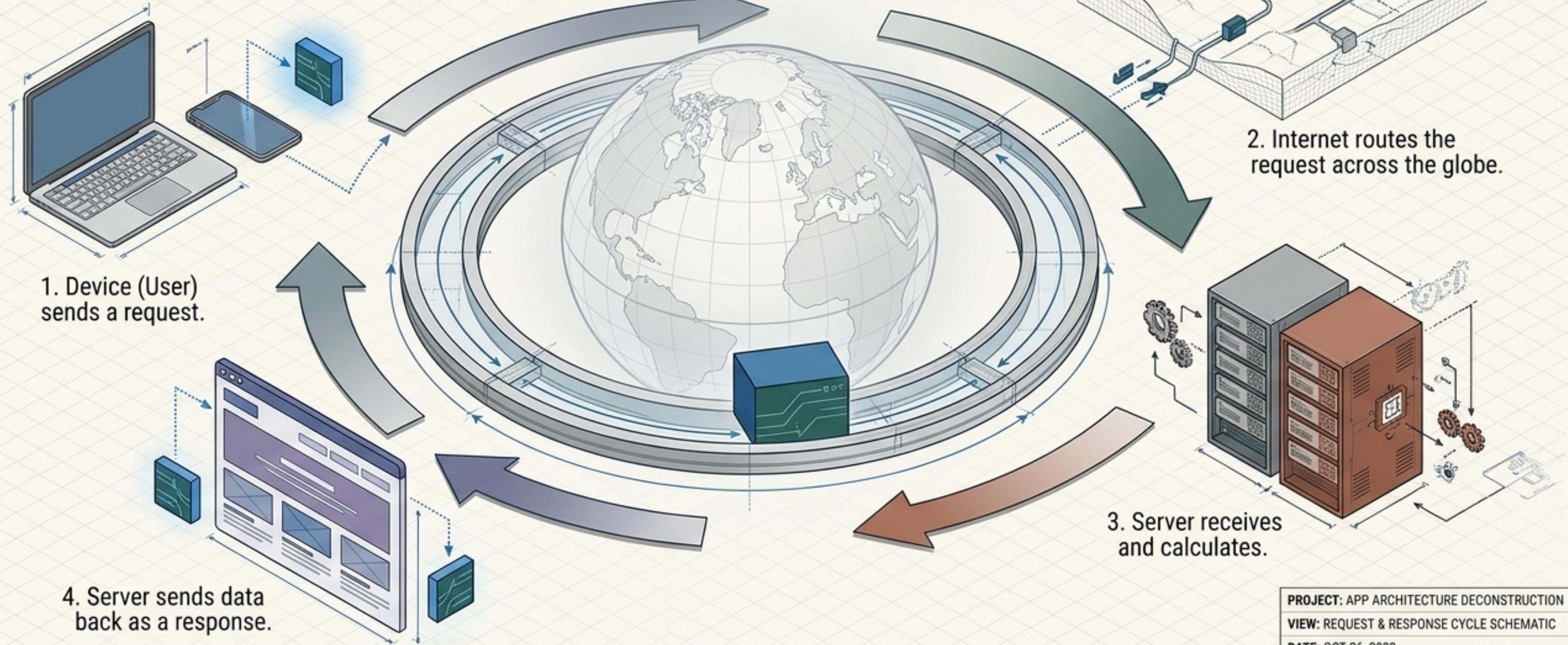
VIEW: EXPLODED ISOMETRIC SCHEMATIC

DATE: OCT 26, 2023

DRAWN BY: ARCHITECTURAL SYSTEMS

The Core Mechanism: Request & Response

Apps are simply global communication systems.
You ask; the system answers.



PROJECT: APP ARCHITECTURE DECONSTRUCTION
VIEW: REQUEST & RESPONSE CYCLE SCHEMATIC
DATE: OCT 26, 2023
DRAWN BY: ARCHITECTURAL SYSTEMS

Translating the Digital Workflow

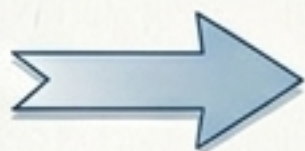
PHYSICAL ANALOGY



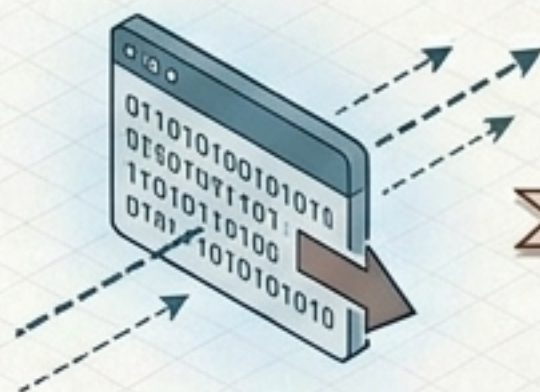
Customer
(Sitting at table)



User
(Clicking a button)



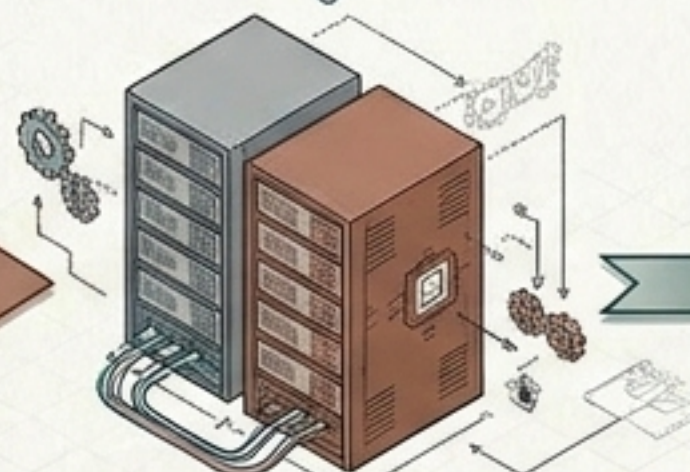
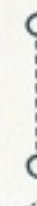
Waiter
(Taking the order)



Internet Request
(Message in transit)



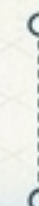
Kitchen
(Preparing food)



Server
(Processing the logic)



Meal
(Delivered to table)



Website Response
(The loaded page)

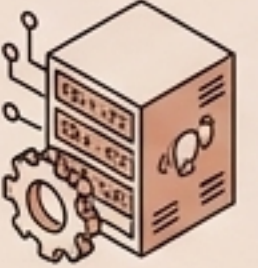
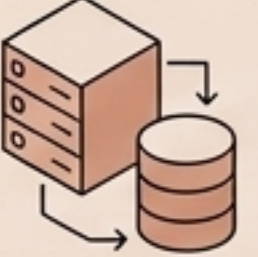
PROJECT: APP ARCHITECTURE DECONSTRUCTION

VIEW: WORKFLOW TRANSLATION SCHEMATIC

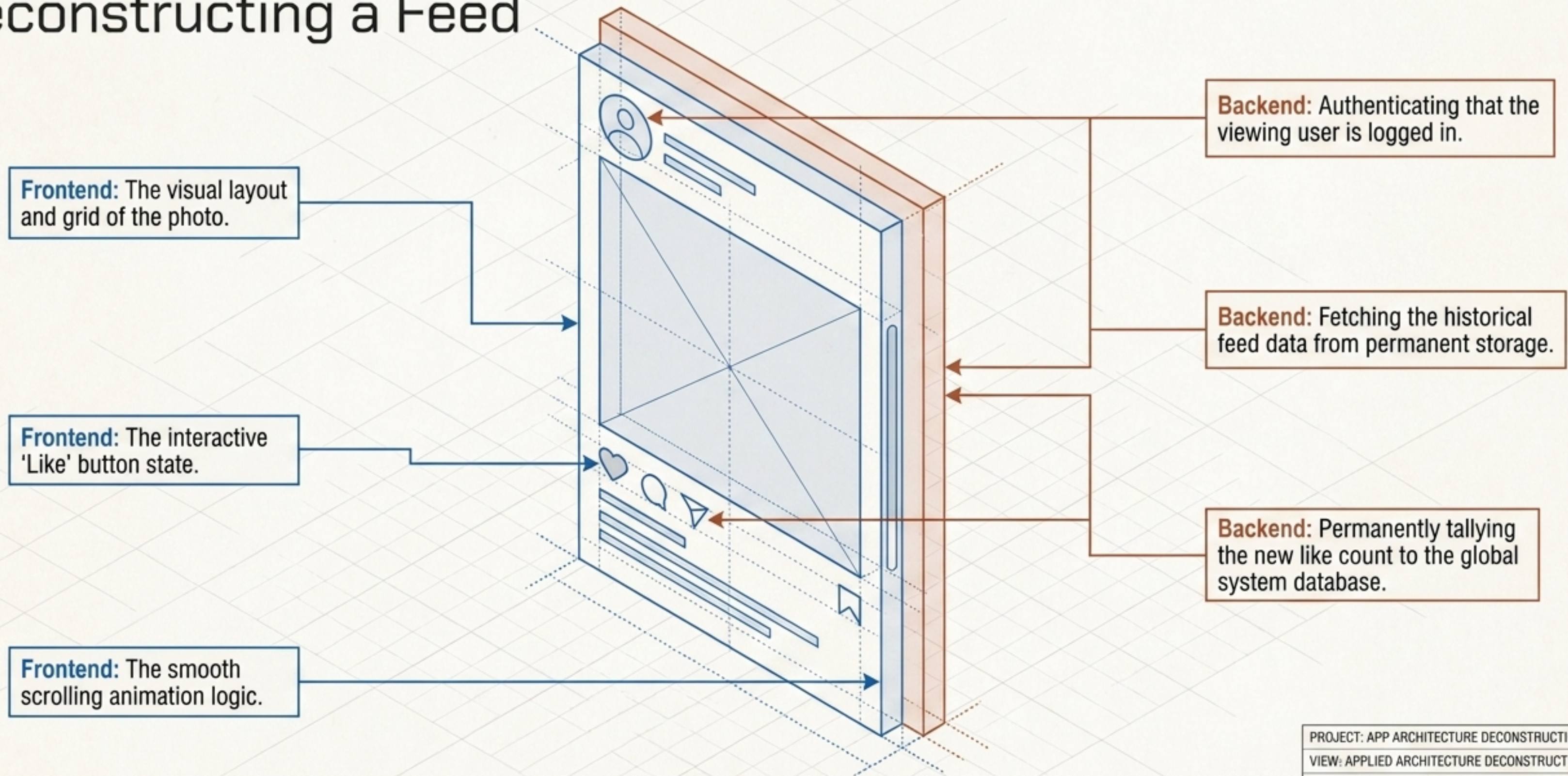
DATE: OCT 26, 2023

DRAWN BY: ARCHITECTURAL SYSTEMS

The Great Divide: Frontend vs Backend

	FRONTEND	BACKEND
Definition	What users see and touch. 	What powers the system behind the scenes. 
Primary Role	Interfaces, layouts, and interactive visual elements. 	System logic, security authentication, and data management. 
Core Components	Buttons, menus, animations, typography, and responsive layouts. 	Server logic engines, permanent databases, and API connectors. 

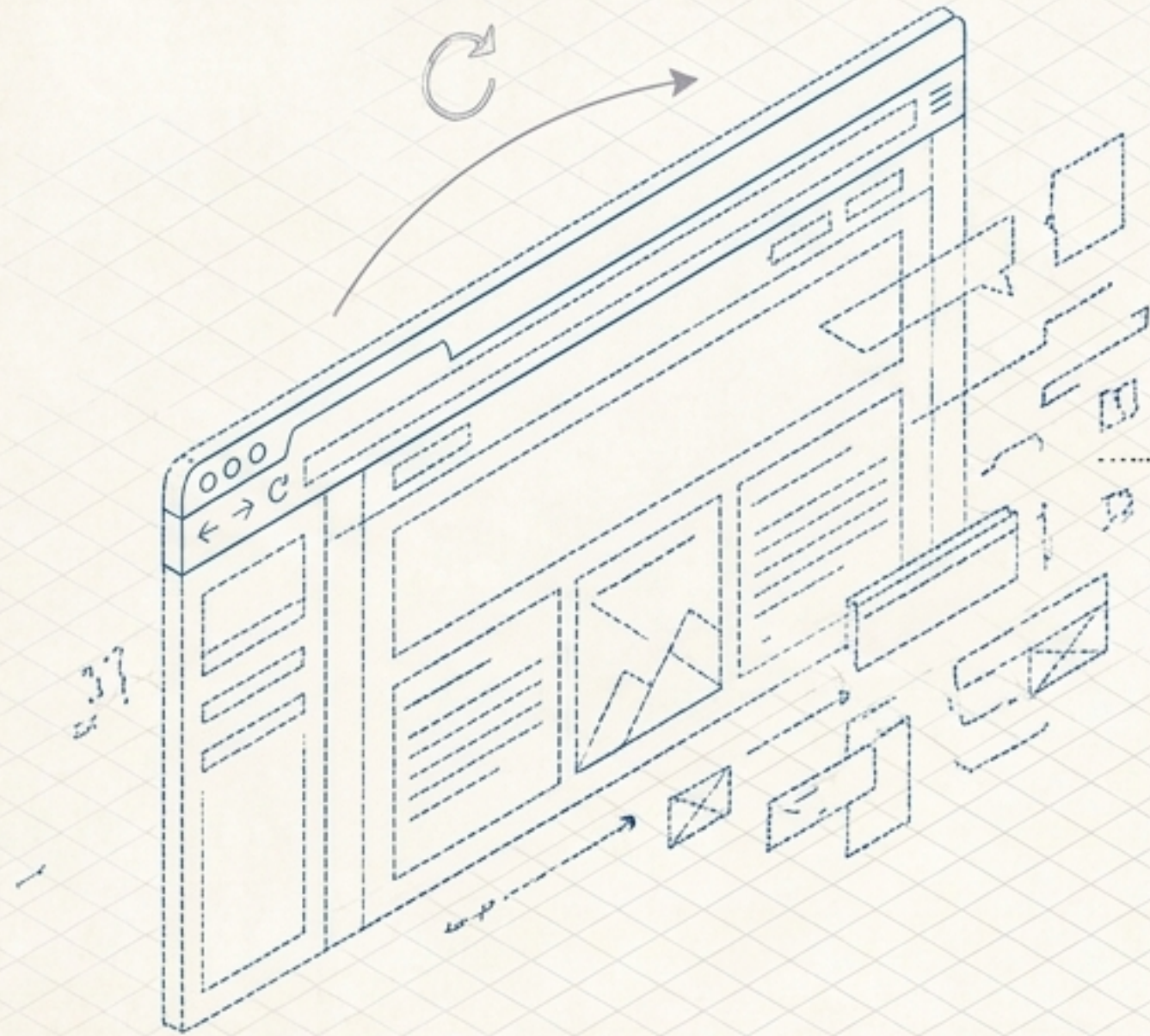
Applied Architecture: Deconstructing a Feed



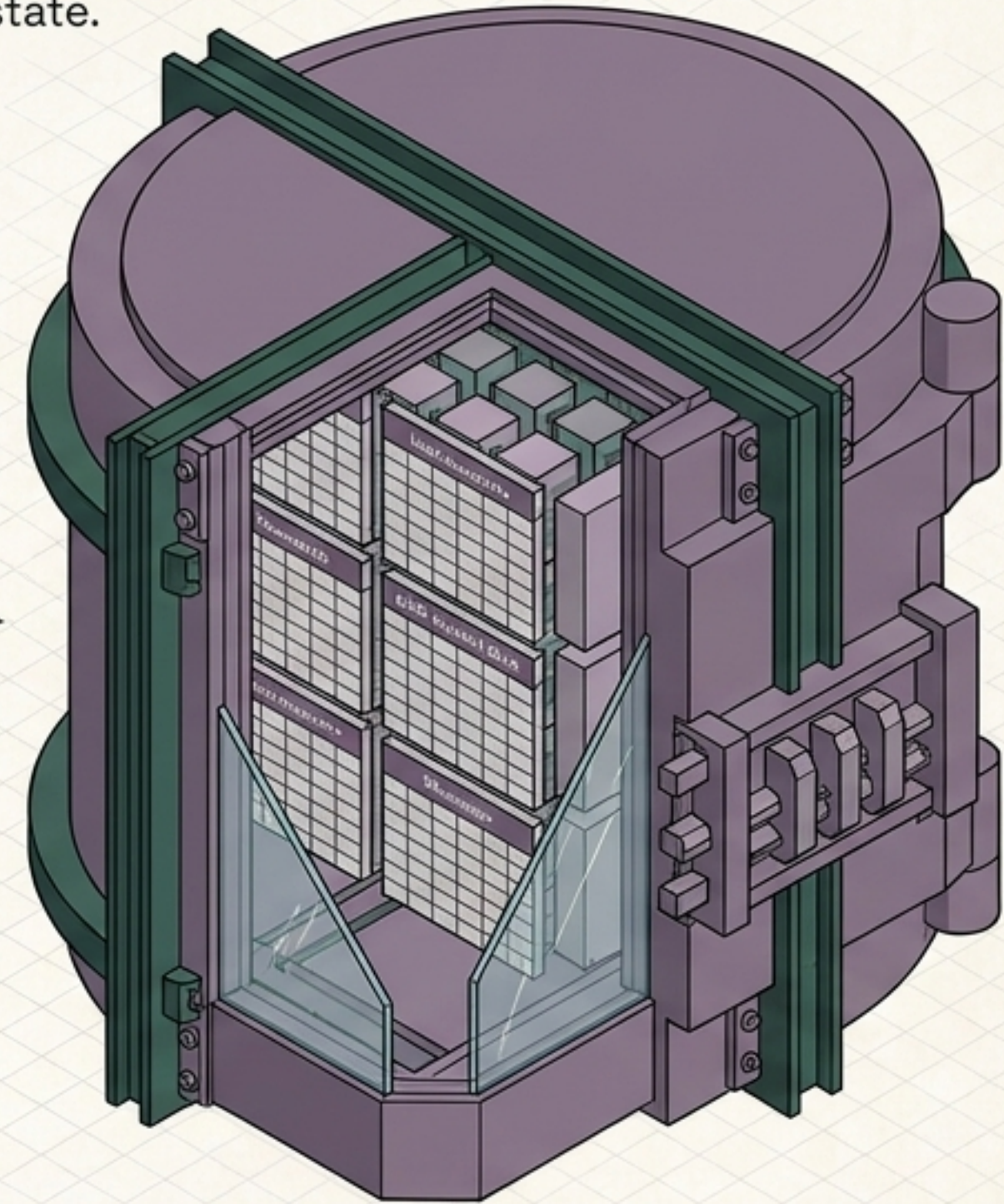
PROJECT: APP ARCHITECTURE DECONSTRUCTION
VIEW: APPLIED ARCHITECTURE DECONSTRUCTION
DATE: OCT 26, 2023
DRAWN BY: ARCHITECTURAL SYSTEMS

Permanent Storage: The Database

Without a database, all information disappears the moment you refresh the page. Databases act as secure records rooms that lock information into a permanent state.



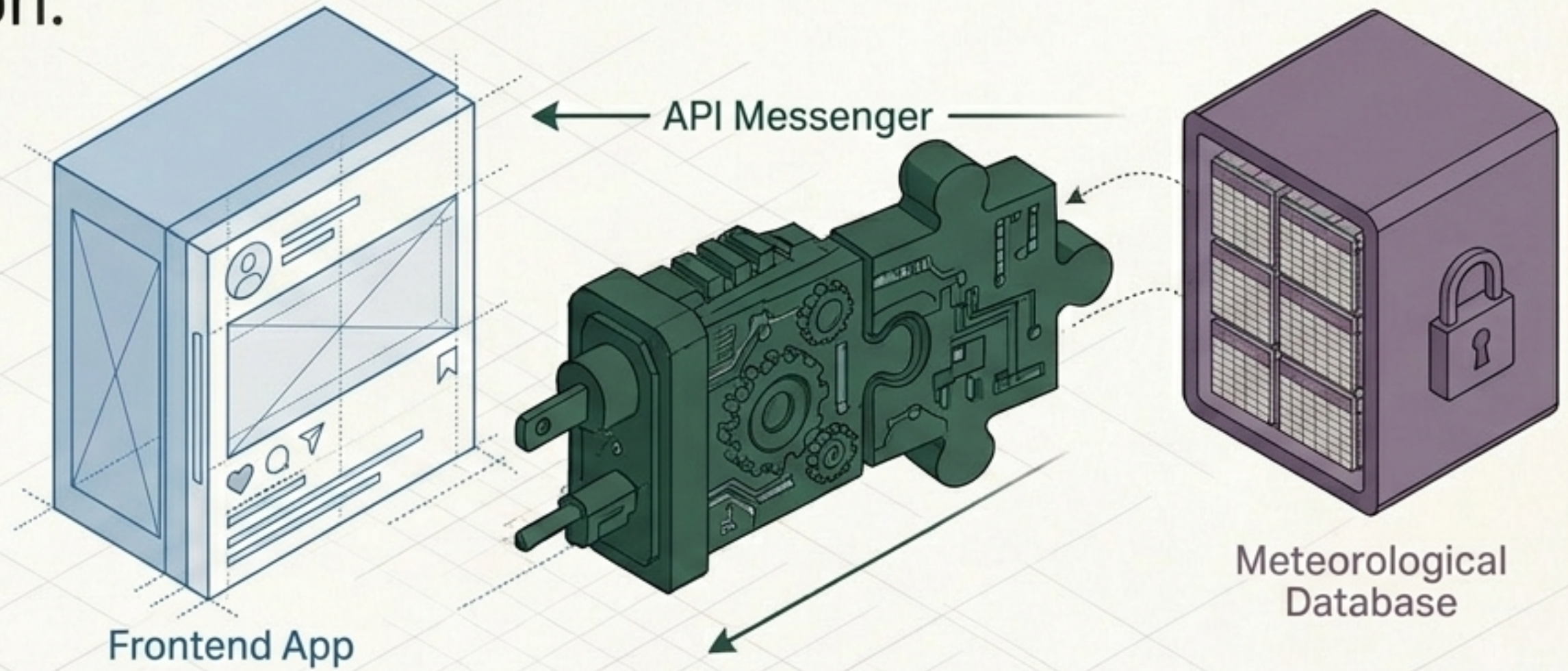
Volatile Memory



Persistent Database

System Communication: What are APIs?

Modern apps are rarely built from scratch. They are assembled by connecting multiple APIs to borrow functionality from external systems.



Identity API



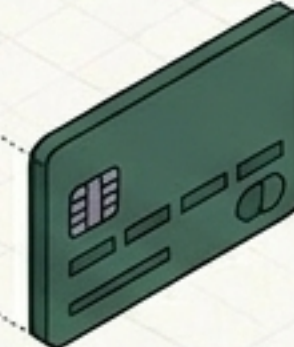
Mapping API



Mapping API



Payment API



Payment API

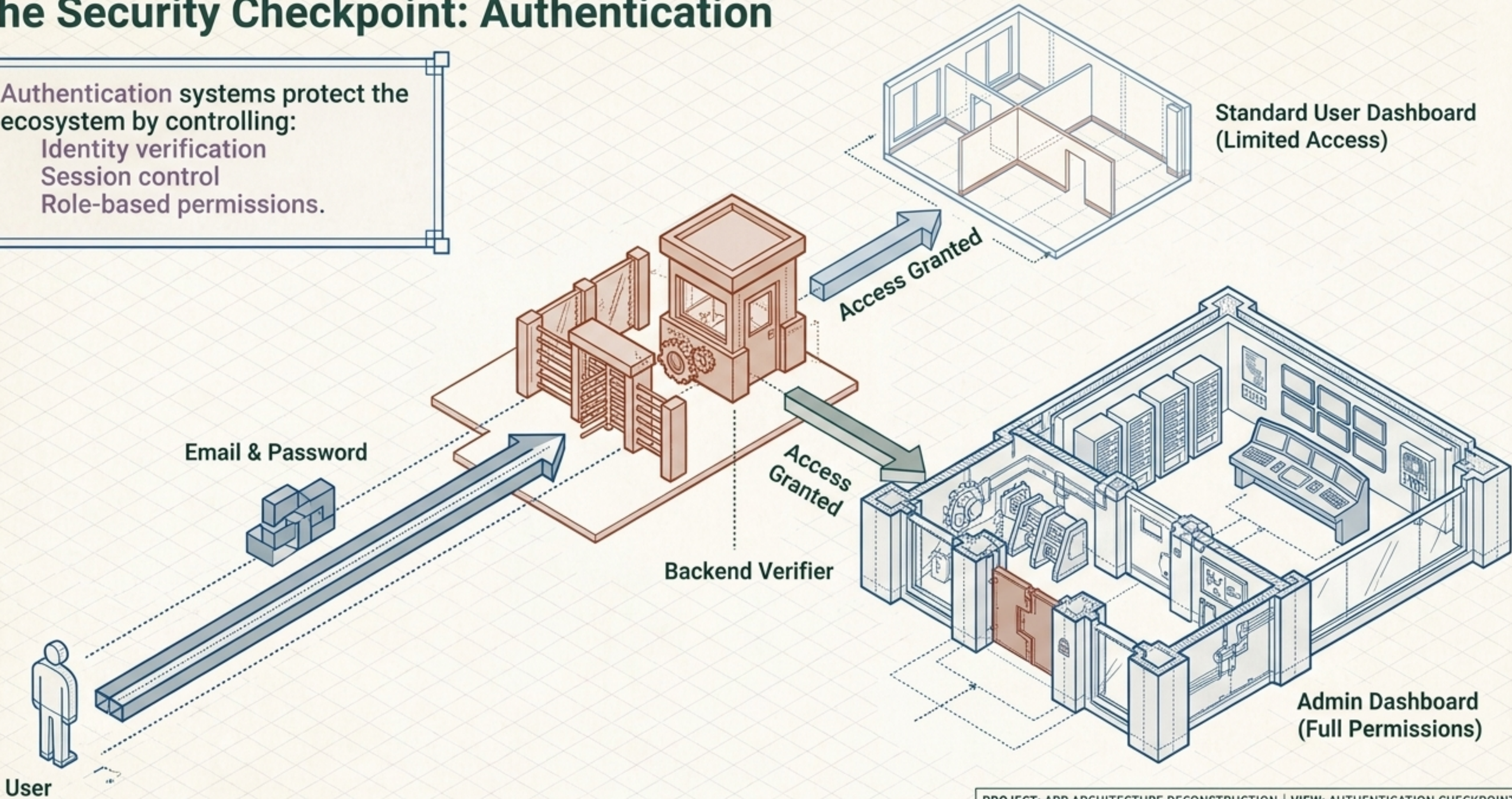


AI Integration API

The Security Checkpoint: Authentication

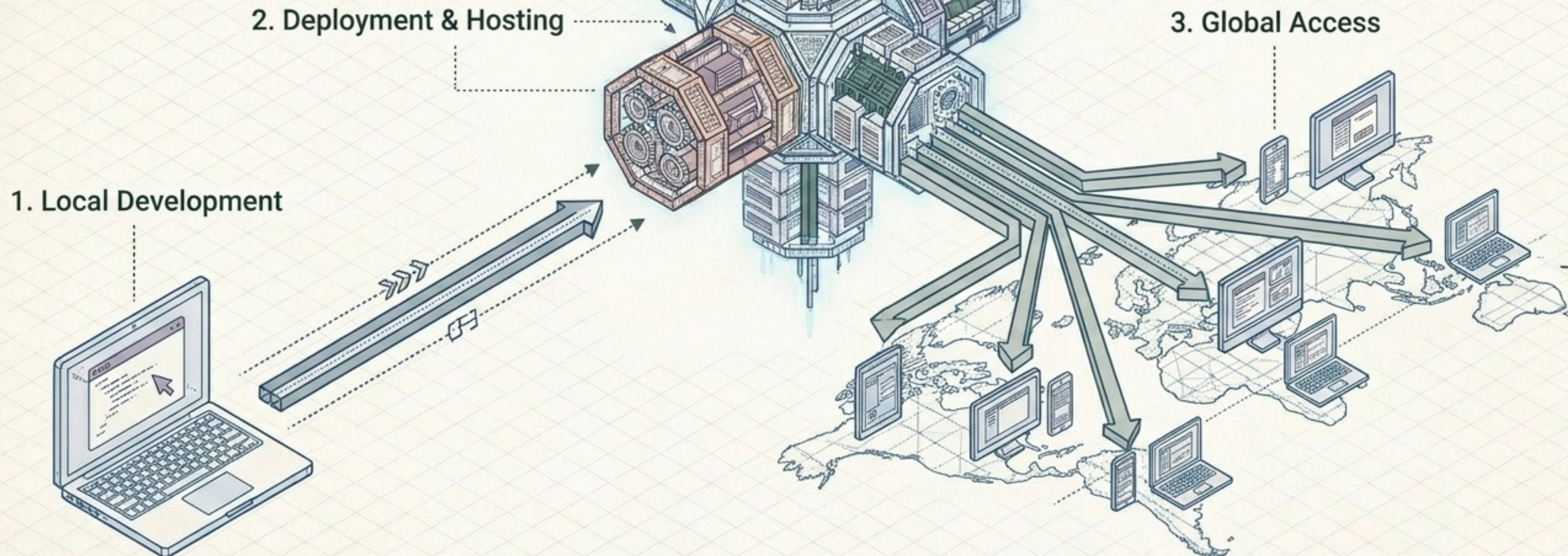
Authentication systems protect the ecosystem by controlling:

- Identity verification
- Session control
- Role-based permissions.



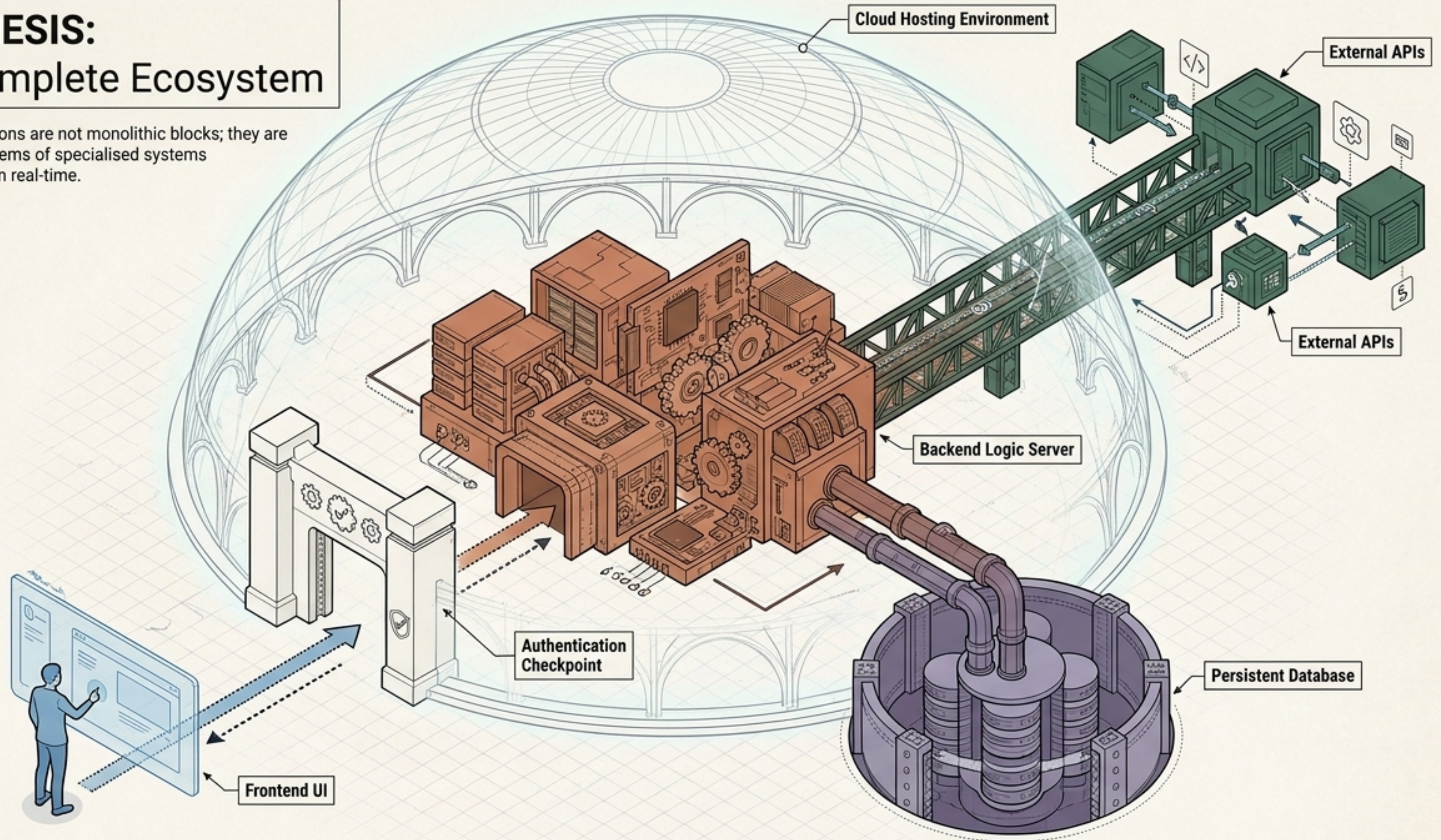
Going Global: Cloud Hosting

Hosting systems take code from a single developer's machine and place it on internet-connected servers, making the application accessible worldwide.



SYNTHESIS: The Complete Ecosystem

Modern applications are not monolithic blocks; they are dynamic ecosystems of specialised systems communicating in real-time.



5 Pillars of Web Architecture

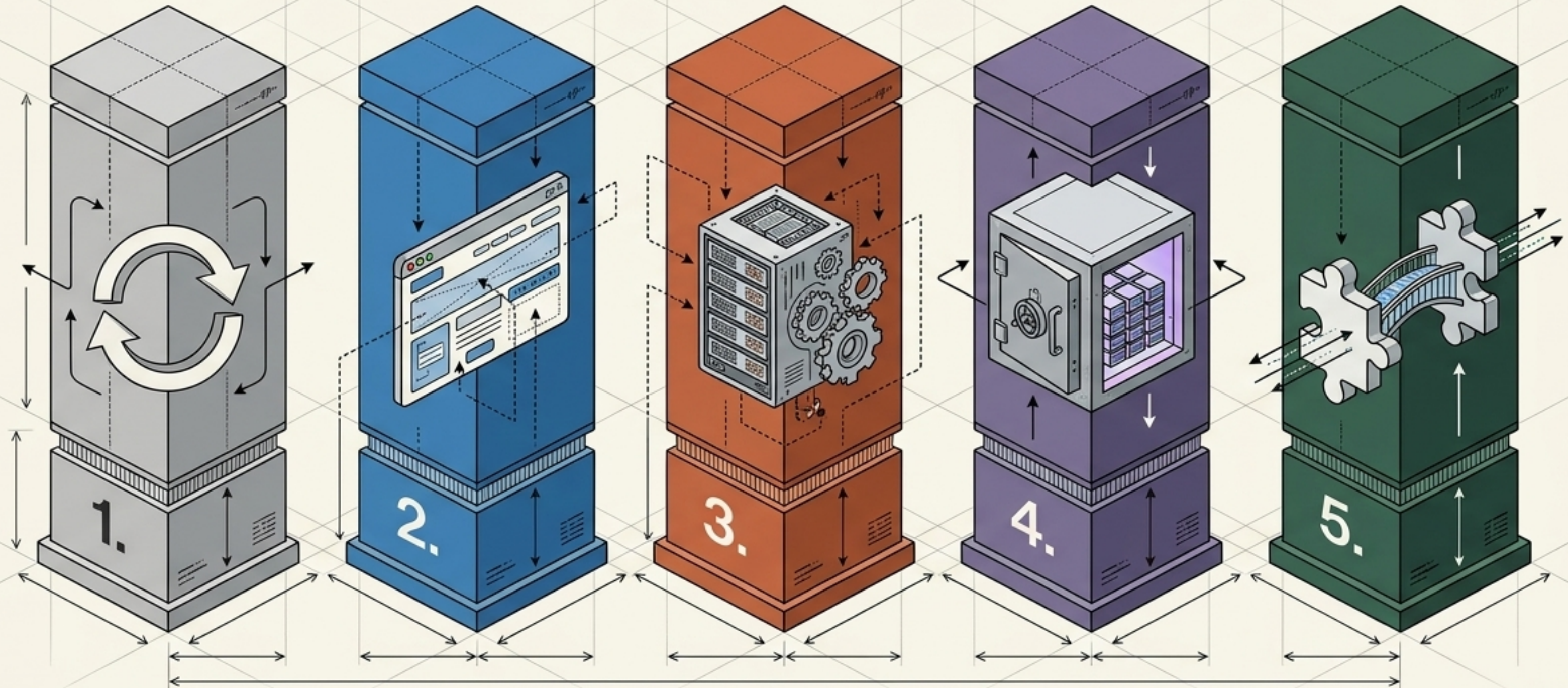
Requests & Responses:
The communication loop
powering the web.

Frontend:
The visual interface
users interact with.

Backend:
The hidden logic
powering the system.

Databases:
The permanent storage
for application data.

APIs:
The messengers allowing
distinct systems to collaborate.



What's Next?

